# Dynamic Correlation Regimes in Major Commodity Futures Markets

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**Abstract**

We analyze daily log returns for 5 commodity futures contracts from 2010-01-05 to 2023-12-29, covering 3514 synchronized trading observations. The sample includes energy, metals, and agricultural markets, enabling cross-sector comparison of return distributions and dependence structure. Correlation evidence indicates meaningful cross-commodity heterogeneity. The highest average rolling correlation is observed for Crude Oil Wti / Copper (mean rolling correlation 0.336; mean DCC correlation 0.343), while the weakest pair is Gold / Natural Gas (mean rolling correlation 0.010; mean DCC correlation 0.012). This suggests that market integration is concentrated in specific pair linkages rather than uniform across the panel.

## 1 Introduction

Understanding time-varying cross-commodity dependence is central to portfolio construction, risk budgeting, and stress-testing in futures markets. Correlation structure is not static: macro cycles, policy regimes, and inventory shocks can induce abrupt changes in co-movement. This paper studies dynamic dependence across key commodity contracts and evaluates both smooth evolution and discrete structural shifts.

## 2 Data

We analyze daily log returns for 5 commodity futures contracts from 2010-01-05 to 2023-12-29, covering 3514 synchronized trading observations. The sample includes energy, metals, and agricultural markets, enabling cross-sector comparison of return distributions and dependence structure.

Table 1 reports distributional properties of daily log returns for the commodity panel.

Table 1: Summary statistics for daily commodity log returns.

| Series | Mean | Std | Min | Max | Skew | Kurtosis |
|---|---|---|---|---|---|---|
| Crude Oil Wti | 0.00013 | 0.02623 | -0.28221 | 0.31963 | 0.07591 | 24.85305 |
| Gold | 0.00018 | 0.01012 | -0.09821 | 0.05778 | -0.57344 | 6.09923 |
| Corn | 0.00005 | 0.01798 | -0.26862 | 0.08618 | -1.74582 | 23.19315 |
| Natural Gas | -0.00025 | 0.03438 | -0.30048 | 0.38173 | 0.17607 | 8.40944 |
| Copper | 0.00005 | 0.01426 | -0.07551 | 0.07199 | -0.09360 | 1.95824 |

Table 2: Pairwise correlation diagnostics with regime break counts.

| Pair | Mean Rolling Corr | Regime Breaks | Mean DCC Corr |
|---|---|---|---|
| Corn / Copper | 0.1337 | 40 | 0.1370 |
| Corn / Natural Gas | 0.0654 | 41 | 0.0756 |
| Crude Oil Wti / Copper | 0.3358 | 39 | 0.3430 |
| Crude Oil Wti / Corn | 0.1620 | 39 | 0.1548 |
| Crude Oil Wti / Gold | 0.1526 | 41 | 0.1620 |
| Crude Oil Wti / Natural Gas | 0.1126 | 43 | 0.1103 |
| Gold / Copper | 0.2608 | 39 | 0.2623 |
| Gold / Corn | 0.0821 | 41 | 0.0817 |
| Gold / Natural Gas | 0.0096 | 43 | 0.0117 |
| Natural Gas / Copper | 0.0507 | 43 | 0.0595 |

# 3 Methodology

We estimate dependence with three complementary approaches. First, we compute rolling 252-day pairwise correlations to capture medium-horizon variation in co-movement. Second, we apply structural break detection using the PELT algorithm with an RBF cost and quarterly minimum segment length to identify regime boundaries. Third, we estimate DCC-GARCH(1,1) dynamics by fitting univariate GARCH(1,1) models to each return series, extracting standardized residuals, and recursively estimating the dynamic conditional correlation process.

# 4 Results

Table 2 summarizes average rolling correlations, break counts, and DCC mean correlations.

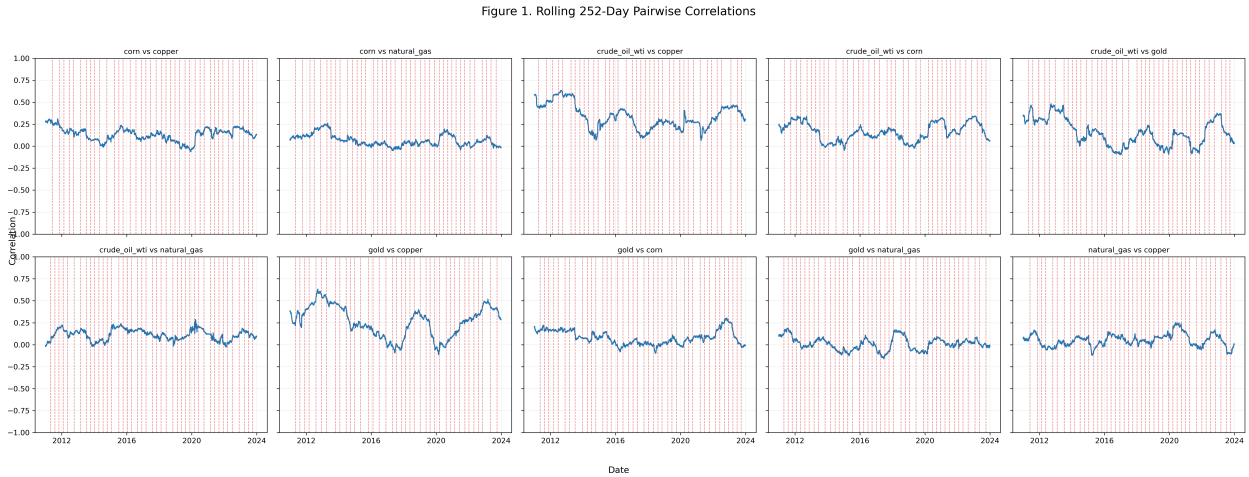


Figure 1: Rolling 252-day pairwise correlations with detected break dates.

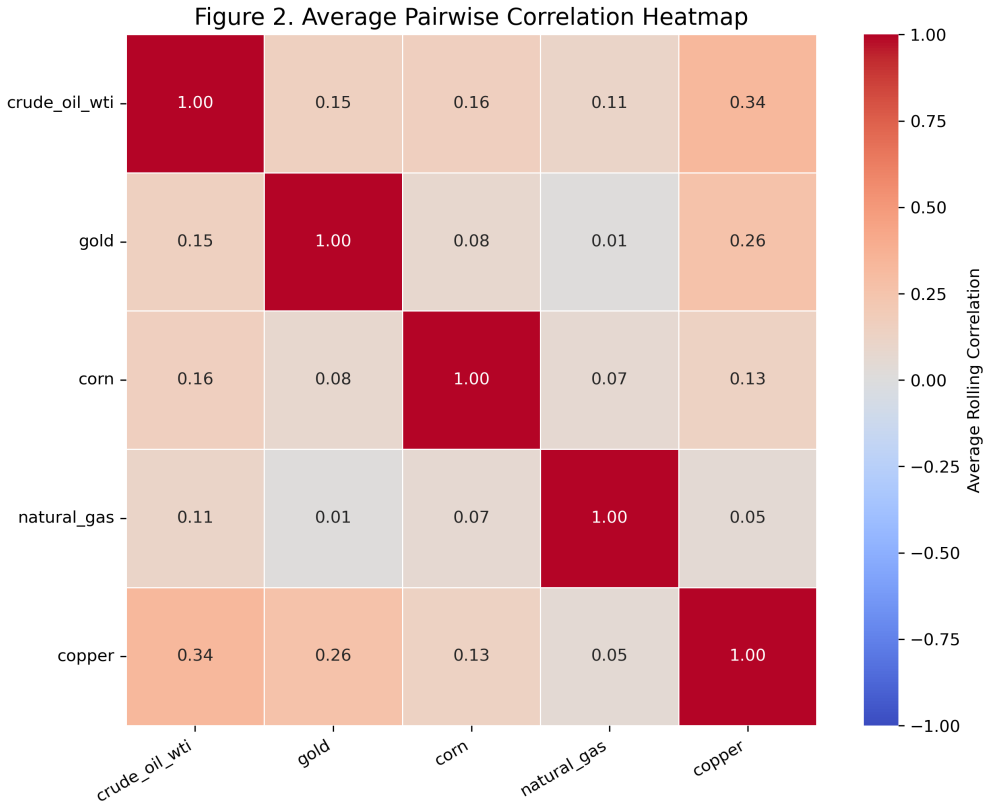


Figure 2: Average pairwise correlation heatmap.

Correlation evidence indicates meaningful cross-commodity heterogeneity. The highest average rolling correlation is observed for Crude Oil Wti / Copper (mean rolling correlation 0.336; mean DCC correlation 0.343), while the weakest pair is Gold / Natural Gas (mean rolling correlation 0.010; mean DCC correlation 0.012). This suggests that market integration is concentrated in specific pair linkages rather than uniform across the panel.

## 5 Conclusion

Regime diagnostics reveal persistent instability in correlation dynamics: pairs exhibit an average of 40.9 detected breakpoints, with the highest instability in Crude Oil Wti / Natural Gas (43 breaks). These structural shifts imply that static correlation assumptions are likely misspecified over long horizons, and that time-varying dependence models are necessary for robust risk estimation.

## References

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